DSCI 351: Recommender Systems

Course Syllabus

Credits: 3-hour [3 credits]

Spring 2026

General Information

Course Instructor: Lei Wang
Instructor Contact Information: lw474@drexel.edu
Lecture Hours; Location: Tuesdays and Thursdays 10:30-11:50 AM @ 3675 Market St, Room 1056
Office Hours; Location: TBD

Email is the best way to contact. **Be sure to include a subject line and start the subject with a reference to the course.** For example: "DSCI 351 Question" would work. **Email without a clear subject may be deleted with spam**

Student Learning Information

Course Description

Recommender systems are electronic commerce information filtering systems to predict items that the users may have interest in. The goal of this course is to provide an overview of recommender systems, including content-based and collaborative algorithms for recommendation, programming of recommender systems, and evaluation and metrics for recommender systems.

The course introduces all relevant topics of Recommender Systems: overview, collaborative filtering, content-based recommending, knowledge-based systems, recommender system evaluation and advanced topics. Students will gain hands-on experiences with assignments and a term project.

College/Department: College of Computing & Informatics
Repeat Status: Not repeatable for credit
Restrictions: None
Pre-requisites: INFO 212

Course Purpose Within a Program of Study

This course is an elective available to undergraduate students, in particular students in the BS Data Science and BS Information Systems programs.

Recommender systems are both data mining techniques and applications. The course provides a balance between these two perspectives while taking a view of the recommender systems.

Statement of Expected Learning

Upon successful completion of this course, a student will be able to:

- Understand the concepts of recommender systems

- Extract content and utilize the characteristics of items to make recommendation of similar items
- Develop models to capture user behaviors and make recommendations to users who have similar interests
- Evaluate the performance of recommender systems

Course Materials

Required and Recommended Texts, Readings, and Resources

- Charu C. Aggarwal, Recommender Systems: The Textbook (RST), Springer, 2016. ISBN: 9783319296579
- Jannach, D., Zanker, M., Felfernig, A., & Friedrich, G, Recommender Systems: An Introduction (RSI), Cambridge University Press, 2010. ISBN: 0521493366

Recommended textbooks are available to Drexel students through the University Libraries.

Required and Supplemental Materials and Technologies

Students are expected to have a Python programming environment configured, with Jupyter notebooks, Scikit-Learn, Numpy, Pandas, and Matplotlib installed by the following by the start of the first week. Additional resources include

- Python's docs
- Scipy docs
- Scikit-learn docs
- iPython docs

Assignments, Assessments, and Evaluations

Graded Assignments and Learning Activities

Group assignments will be distributed to provide hands-on experience in different types of recommender systems, covering the following topics:

- Memory-based collaborative-filtering recommender system
- Model-based collaborative-filtering recommender system
- Content-based recommender system

Group project: One open-ended group project

1. Type of the recommender system
2. Methods Implemented
3. Evaluation, and Interpretation, with Documentation

Individual quizzes: there will be quizzes to be completed individually on Bb Learn.

The term project is open-ended and sophisticated in a way that students can select key problems, explore data, and build a recommender system based on students' own interests.

Submitting Assignments

- You must submit assignments electronically via Blackboard Learn no later than 11:59 pm on the due day.
- No email submission accepted.
- The name(s) and student ID(s) of all submitters of the assignment along with the course number and assignment number must be clearly printed on the first page and file name.
- University rules and policies regarding academic honesty are followed to the letter.

Late or Missing Work

Each item of course work you turn in has a due date and time:

- Deliverables are due at the time and date indicated by instructions. Written assignments are due no later than 11:59 pm on the due day.
- There will be a 10% (absolute value) deduction for each day of lateness, to a maximum of 3 days; assignments will not be accepted beyond that point.
- Missing work will earn a zero grade.

Re-Marking

- If you are dissatisfied with a grade or point deduction, you can request re-marking (must within 5 days of receiving grade).
- All re-marking requests must be done through written (paper or email) descriptions of why you think the grade is in error.
- Please note that it is very rare that changing disputed grades on assignments or quizzes actually affects a student's calculated letter course grade. The corrections are usually insignificant. If you wish to appeal a course grade, school policies apply.

Teams for Assignment and Project

The assignments and project are team effort.

- Each team can have up to 3 members. Student is allowed to work alone in a one-person group.
- Each team (include team with single member) must self-enroll on Bb Learn before the end of the first week. Members are encouraged to stay in the same team through the entire quarter. If team members change, a written request must be submitted to the instructor.
- Each team submits a single completed assignment for each assignment and a single project zip file for the final project.
- Each team will be asked to present the project at the end of the term.

Team Member Evaluation

- All students on a single team initially receive the same grade. However, each team member will evaluate the performance of every other member in his/her team. The instructor reserves the right to adjust the grade based on the team evaluation results.
- All students of a team must have their names listed clearly on the front page of the reports. Each team's work must be unique – one team cannot collaborate with another team.

Grading Matrix

Students will not receive letter grades for individual assignments. Grades are calculated as:

Project:	35% (proposal 10%, implementation 25%)
Homework:	45%
Quiz:	20%
Total:	100%

*Possibly bonus points for completing the course evaluation and provide screenshot of the completion page.

Grade Scale

The following scale will be used to convert points to letter grades:

Points	*Grade*	*Points*	*Grade*	*Points*	*Grade*
97-100	A+	84-86.99	B	70-73.99	C-
94-96.99	A	80-83.99	B-	67-69.99	D+
90-93.99	A-	77-79.99	C+	64-66.99	D
87-89.99	B+	74-76.99	C	0-63.99	F

Note that the instructor may revise this conversion if/when necessary.

Course Schedule

This course will be taught in person through the Drexel University College of Computing & Informatics online Blackboard at http://learn.dcollege.net. If you have any questions about, or problems with Blackboard, please call (215) 895 1224 or visit the support portal by going to http://www.drexel.edu/irt/help/a-z/learn/ at any time.

It is expected that students taking this course will spend a minimum of 10 hours per week on class work, therefore it is important to pay attention to time management in order to complete the course successfully. Although you can access the course materials online at any time, plan to set aside time each week to complete the assignments and projects. Please check the Blackboard site regularly for announcements, assignments, projects, and lecture notes etc.

The course schedule is as follows:

Week	Topics	Readings	Deliverables
Week 1	Introduction to Recommender Systems Overview of different systems	RSI: chap 1 RST: chap 1	
Week 2	User-based and item-based collaborative filtering Python coding demo (memory-based)	RSI: chap 2 RST: chap 2	
Week 3	Data mining methods for collaborative filtering	RSI: chap 2 RST: chap 3	Assignment 1
Week 4	Data mining methods (cont.) Model-based collaborative filtering Python coding demo (model-based)	RSI: chap 2 RST: chap 3	
Week 5	Feature extraction for content-based systems Feature selection	RSI: chap 3 RST: chap 4	Project proposal
Week 6	Learn User Profile Python coding demo (content-based)	RSI: chap 3 RST: chap 4	Assignment 2
Week 7	Simple recommender Evaluating recommender system	RSI: chap 3 RST: chap 4	
Week 8	Evaluating recommender system Constraint-based systems (knowledge-based)	RSI: chap 7 RST: chap 7	Assignment 3
Week 9	Case-based systems (knowledge-based) Python coding demo (knowledge-based)	RSI: chap 4 RST: chap 5	
Week 10/ Finals week	Project presentation		Project

Notice: Appropriate Use of Course Materials

It is important to recognize that some or all of the course materials provided to you may be the intellectual property of Drexel University, the course instructor, or others. Use of this intellectual property is governed by Drexel University policies, including the policy found here: https://drexel.edu/it/about/policies/policies/01-Acceptable-Use/

Briefly, this policy states that course materials, including recordings, provided by the course instructor may not be copied, reproduced, distributed or re-posted. Doing so may be considered a breach of this policy and will be investigated and addressed as possible academic dishonesty, among other potential violations. Improper use of such materials may also constitute a violation of the University's Code of Conduct found here: https://drexel.edu/compliance-privacy-audit/compliance/policies/cpo-1-01/ *and will be investigated as such.*

Academic Policies

This course follows university, college, and department policies, including but not limited to:

- Academic Honesty: http://www.drexel.edu/provost/policies/academic_dishonesty.asp
  - You may use AI programs e.g. ChatGPT to help generate ideas and brainstorm. However, you should note that the material generated by these programs may be inaccurate, incomplete,

or otherwise problematic. Beware that use may also stifle your own independent thinking and creativity.

- You may not submit any work generated by an AI program as your own. If you include material generated by an AI program, it should be cited like any other reference material (with due consideration for the quality of the reference, which may be poor).
- Any plagiarism or other form of cheating will be dealt with severely under relevant Drexel policies.

- Student Life Honesty Policy from Judicial Affairs: http://www.drexel.edu/provost/policies/academicintegrity
- Students with Disability Statement: http://drexel.edu/oed/disabilityResources/faculty/SyllabusStatement/
- Course Drop Policy: http://www.drexel.edu/provost/policies/course_drop.asp
- Academic Integrity: https://drexel.edu/provost/policies/academic-integrity/
- Academic Integrity Policy: https://drexel.edu/studentlife/community_standards/code-of-conduct/academic-integrity-policy/
- Drexel Student Learning Priorities: http://drexel.edu/provost/assessment/outcomes/dslp/
- Office of Disability Resources: http://www.drexel.edu/ods/student_reg.html

The instructor(s) may, at his/her/their discretion, change any part of the course before or during the term, including assignments, grade breakdowns, due dates, and schedule. Such changes will be communicated to students via the course web site. This web site should be checked regularly and frequently for such changes and announcements.

Students requesting accommodations due to a disability at Drexel University need to request a current Accommodations Verification Letter (AVL) in the ClockWork database before accommodations can be made. These requests are received by Disability Resources (DR), who then issues the AVL to the appropriate contacts. For additional information, visit the DR website at drexel.edu/oed/disabilityResources/overview/, or contact DR for more information by phone at 215.895.1401, or by email at disability@drexel.edu.